

E07 Adiabatic Batch Reactor Example

The irreversible, elementary liquid phase reaction $A + B \rightarrow C$ is carried out adiabatically in a 10-L batch reactor. Species A and B are present initially in equimolar amounts. Determine the time needed to achieve 90% conversion and the resulting temperature. $T_0 = 300 \text{ K}$.

$$C_{A0} = 0.1 \text{ mol/L}$$

$$C_{PA} = C_{PB} = 0.015 \text{ kcal/mol/K}$$

$$C_{PC} = 0.030 \text{ kcal/mol/K}$$

$$\Delta H_{f,A}^0 = -20 \text{ kcal/mol}$$

$$\Delta H_{f,B}^0 = -15 \text{ kcal/mol}$$

$$\Delta H_{f,C}^0 = -41 \text{ kcal/mol}$$

$$E_{act} = 10 \text{ kcal/mol}$$

$$k = 0.01 \text{ L/mol/s at } 325 \text{ K}$$

Decision Guide: Adiabatic Batch Reactor Example

You are asked to determine:

1. The time required to achieve 90% conversion.
2. The reactor temperature at 90% conversion.

Step 1: Identify What Kind of Problem This Is

Ask yourself:

What reactor?

→ Batch Reactor

Is it isothermal?

→ No

Is it adiabatic?

→ Yes

Because the reactor is non-isothermal and adiabatic, both a mole balance and an energy balance will be required.

Step 2: Write the Batch Reactor Mole Balance

Use the Batch Reactor Mole Balance:

$$N_{A0} \frac{dX}{dt} = -r_A V: \text{CBE 3610 Reactor Design Handbook Equation (20)}$$

Keep this equation available—we will return to it after developing the rate expression.

Step 3: Determine the Heat of Reaction

Calculate:

ΔH_{rxn} from the heats of formation provided in the problem statement.

Use:

$$\Delta H_{\text{rxn}}^{\circ} = \left(\frac{d}{a} H_D^{\circ} + \frac{c}{a} H_C^{\circ} - \frac{b}{a} H_B^{\circ} - H_A^{\circ} \right)$$

CBE 3610 Reactor Design Handbook Equation (25)

Step 4: Write the Adiabatic Energy Balance

Because the reactor is adiabatic: $Q = 0$

Use the adiabatic energy balance to write temperature as a function of conversion:

$$T = T_0 + \frac{(-\Delta H_{\text{rxn}}^{\circ})X}{\sum_i \Theta_i C_{Pi}}$$

CBE 3610 Reactor Design Handbook Equation (27)

This gives: $T = f(X)$ which is usually the easiest relationship to obtain first.

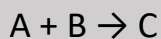
Step 5: Write the Rate Law

The problem states that the reaction is:

Irreversible

Elementary

For the reaction:



the rate law is:

$$-r_A = k C_A C_B$$

using the general form:

$$-r_A = [k(T)] \cdot [f(C_A, C_B)]$$

CBE 3610 Reactor Design Handbook Equations (44) and (45)

Step 6: Calculate the Temperature-Dependent Rate Constant

The problem provides:

- Activation energy

- Reference temperature
- Reference rate constant

Use:

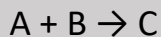
$$k(T) = k(T_{ref}) \left[\frac{E}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right]$$

CBE 3610 Reactor Design Handbook Equation (49)

Substitute the temperature relationship obtained from Equation (27).

Step 7: Express Concentrations in Terms of Conversion

For:



and constant-volume liquid-phase conditions:

$$C_A = C_{A0}(1 - X)$$

$$C_B = C_{B0}(1 - X)$$

using stoichiometry and the definition of conversion.

Step 8: Develop a Rate Expression Containing Only X

Combine:

- Equation (27)
- Equation (49)
- Equation (44)
- Equation (45)

to obtain:

$-r_A = f(X)$ only. At this point: Temperature has been eliminated because $T = f(X)$.

Step 9: Substitute into the Batch Reactor Mole Balance

Return to:

$$N_{A0} \frac{dX}{dt} = -r_A V$$

CBE 3610 Reactor Design Handbook Equation (20)

Substitute your expression for:

$-r_A = f(X)$ to obtain: $\frac{dX}{dt} = f(X)$

Step 10: Integrate to Find Time

Rearrange the mole balance:

$$dt = \frac{N_{A0}}{-r_A V} dX$$

and integrate from: $X = 0$ to $X = 0.90$ to obtain the required batch time.

Final Answer Checklist

Before submitting your solution, verify that you have:

- ☒ Calculated ΔH_{rxn} using Equation (25)
- ☒ Determined $T(X)$ using Equation (27)
- ☒ Calculated $k(T)$ using Equation (49)
- ☒ Written the rate law using Equations (44)–(45)
- ☒ Used the Batch Reactor Mole Balance, Equation (20)
- ☒ Determined the time required for 90% conversion
- ☒ Determined the final reactor temperature

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (20) Batch Reactor Mole Balance

Eq. (25) Heat of Reaction

Eq. (27) Adiabatic Energy Balance
(Temperature as a Function of Conversion)

Eq. (44) General Rate Law

Eq. (45) Power-Law Kinetics

Eq. (49) Arrhenius Temperature Dependence

Key Insight

Eq. (27)



$T(X)$

Eq. (49)



$k(T)$

Eq. (44)-(45)



$-r_A(X)$

Eq. (20)



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This example demonstrates how the **energy balance, rate law, Arrhenius equation, and batch-reactor mole balance** work together to solve a **non-isothermal reactor-design problem**.